Title: Identifying Spatial and Socio-Economic Drivers and impact of Gentrification in Buffalo (2016–2024)

Sazzad Hossain

GEO 479-559

1.Introduction:

Many American post-industrial cities undergo a long period of economic decline, population loss and neighborhood disinvestment. However, in recent years some of these cities start to experience reinvestment and rising housing demand(Swanstrom et al., n.d.). Buffalo, New York, is an important example of this transformation over time. Buffalo is known as a declining Rust Belt city, which is recently experiencing reinvestment, neighborhood redevelopment and rising housing price in selected area(Adelman et al., 2019a). These changes are improving urban infrastructure and attracting new investment but at the same time it also raising concerns about gentrification and potential displacement of long-term residents.

Before beginning this research, an initial assumption was that population growth and declining crime rates might explain housing price increases in Buffalo neighborhoods. However, preliminary correlation analysis result does not support that assumption. Analysis showed that population density changes and crime reduction alone were not statistically significant predictors of increasing housing values. This motivated a broader investigation into the socio-economic and spatial drivers of neighborhood transformation.

Gentrification is a complex urban process. It involves reinvestment, rising property values, demographic shifts, and changes in neighborhood social structure(Freeman, 2005). Gentrification can occur unevenly across space. Some neighborhood may experience rapid transformation while

others may continue to go through disinvestment(Lees et al.). To know where gentrification in happening or area that is vulnerable for gentrification is important for urban planner or people work for housing policy, and social equity.

2. Objective:

The primary objectives of this project are:

1. To identify census tracts in Buffalo that were vulnerable or vulnerable for gentrification.
2. To identify the socio-economic and spatial factors associated with gentrification.
3. To measure the intensity of gentrification across vulnerable census tracts.
4. Measure the impact of gentrification.

3. Significance:

This research has both social and methodological significance. The findings will help planners and policymakers identify vulnerable tracts that experiencing rapid neighborhood change and potential displacement pressures. The study will help understand how urban redevelopment affects neighborhood housing affordability and demography in Buffalo.

Methodologically, the project demonstrates how GIS, census data, statistical modeling, and spatial analysis can be integrated to study urban transformation. Instead of using a simple yes-or-no classification of gentrification, this project introduces a continuous gentrification intensity framework using standardized Z-scores. This approach allows the study to intensity of neighborhood change and provides a more nuanced understanding of urban transformation.

4. Literature Review

Gentrification is generally known as a process by which economically declining urban neighborhoods experience reinvestment, increasing property values, and demographic change. In other word gentrification as the process by which neighborhoods that previously experienced disinvestment undergoes economic revitalization accompanied by the movement of middle- and upper-income populations into these areas(Freeman, 2005). Sociologist Nail Smith (1979) introduced the Rent Gap Theory to explain why gentrification occurs spatially. According to his theory, gentrification happen when there is a gap emerge between a properties current value and its potential future value(Smith, 1979). So, gentrification happens in place where there is profit-making opportunity arise. This theory suggests that not individual lifestyle preference, it is economic opportunity that act as primary force for gentrification. Gentrification can be triggered by tourism as well. Emerging demand for Airbnb (short time stays) in tourist area is accelerating neighborhood transformation(Çobangil & Ateş, 2025). This also support that capital investment plays a major role in urban restructuring.

Urban research on American cities has long been focused on understanding why cities declined and how poverty continued over time (The Truly Disadvantaged (1987), n.d.). However, in recent time some older industrial cities such as Buffalo and St. Louis have started to show signs of recovery or rebound. This rebound does not happen slowly or naturally. Instead, recovery begins when the difference between a neighborhood's current land value and its possible future higher value becomes very large supporting smith's rant gap theory (Swanstrom et al., n.d.-b). Neighborhood that experienced long period of disinvestment are now attracting new investor. Which result in rise in housing demand and population growth. That makes researcher now to

focus on understanding the spatial and social change that shape the decline and recovery. This understanding is important for urban planning policies.

Economic vulnerability is considered the most important precondition for gentrification. Neighborhoods with lower household incomes, lower property values are often more vulnerable to redevelopment pressure. Because historically economically undervalued land creates opportunities for profitable reinvestment (Freeman, 2005b; Smith, 1979). Demographic characteristics also influence neighborhood transformation. Gentrification often produces racial and socio-economic change in neighborhood. Historically marginalized neighborhoods may experience significant demographic turnover as wealthier residents move into previously lower-income areas (Coley & Adelman, 2021). Population density and immigration patterns are also associated with neighborhood transformation. Immigrant communities contribute to neighborhood revitalization of West Side of Buffalo (Adelman et al., 2019). So, demographic changes can occur alongside increasing housing values and investment.

Environmental amenities such as parks and green space often associated with increasing property values and redevelopment. Proximity to green space can increase neighborhood attractiveness and encourage investment (Anguelovski et al., 2018; Lees et al., n.d.-b). However, the influence of environmental amenities may vary depending on cities geography.

Distance to downtown areas is another important factor in increasing housing price. Central urban neighborhoods often experience earlier waves of reinvestment because of accessibility, transportation infrastructure, and employment opportunities (Smith, 1979; Swanstrom et al., n.d.-b). However, the importance of downtown proximity may decline if investment is primarily driven by low land prices rather than accessibility.

Crime rates are also commonly studied in gentrification research. Transformation of neighborhood can alter the local crime pattern (Zhang & Barr, 2024). However, declining crime rate may not trigger gentrification, instead crime reduction may be a consequence of neighborhood transformation and demographic change.

Gentrification may have different from and they are not all the same. Gentrification pattern in expensive cities like New York or Seattle is very different from older industrial cities like Buffalo. In expensive cities gentrification often causes low-income residents to be pushed out by increasing rents and housing price. On the other hand, cities like St. Louis and Buffalo face a different scenario. These cities experienced long periods of population loss, vacant housing, and economic decline (Adelman et al., 2019c). The main issue was that wealthier residents had already left the city, leaving behind concentrated poverty and disinvestment. Here "gentrification" means that neighborhoods begin to recover its people and investment. Gentrification in old- industrial cities may bring positive effect. Arrival of higher-income residents can improve neighborhood conditions, increase local business activity, and create job opportunities for long-term residents (Freeman, 2005). So, one should be very careful between displacement and re-clustering. This distinction is important for creating fair and effective urban policies.

Many traditional studies classify neighborhoods as either gentrified or non-gentrified. However, binary approaches may oversimplify neighborhood transformation because redevelopment occurs at different intensities across space. Recent urban studies increasingly use standardized indices and continuous scores to capture varying levels of neighborhood transformation (Pichardo et al., 2023; Swanstrom et al., n.d.-b; Vergara et al., 2021). Standardized approaches allow researchers to compare neighborhoods across multiple socio-economic

indicators while reducing problems associated with different measurement scales. This project first identifying economically vulnerable census tracts and then creating a continuous gentrification intensity score using standardized economic change indicators. This approach allows for more detailed spatial interpretation of neighborhood transformation.

5. Research Gap:

Although several studies have examined gentrification in Buffalo, limited research has combined spatial analysis, demographic change, environmental variables, and continuous intensity modeling into a single framework. Many studies also focus primarily on housing prices, crime rate without evaluating broader demographic and tax-related impacts.

This project contributes to existing literature by integrating GIS, statistical modeling, and standardized scoring methods to identify both the drivers and impacts of gentrification across Buffalo census tracts.

6. Study Area:

The study area is the City of Buffalo, New York. Buffalo is located in western New York near Lake Erie and it is historically developed as an industrial and transportation center. During the twentieth century, the city experienced major economic decline associated with deindustrialization, suburbanization, and population loss. These processes produced large numbers of vacant properties, declining neighborhoods, and disinvestment across many parts of the city.

Despite this long-term decline, Buffalo has recently experienced localized redevelopment and increasing property investment. Neighborhoods near downtown and along the West Side have shown substantial increases in housing values and redevelopment activity earlier, before

2015(Adelman et al., 2019). Now the present-day transformation is happening on the east side of Buffalo. These changes make Buffalo an important case study for understanding post-industrial urban transformation.

7. Data and Methodology:

This project integrates demographic, housing, tax, environmental, and spatial data to identify and analyze gentrification patterns in Buffalo between 2016 and 2024. GIS and statistical methods were used to identify socio-economic drivers, vulnerable census tracts and to measure gentrification intensity.

7.1 Data Sources:

Several datasets from government and public data sources were used in this project.

7.1.1 Demographic and Housing Data

Demographic and housing variables were collected from the United States Census Bureau's American Community Survey (ACS) 5-year estimates. Variables included:

- Total population
- Median household income
- Median home value
- Median rent
- Race and ethnicity composition
- Foreign-born population
- Housing occupancy
- Population density
- Housing age

- Vacancy rate

These variables were collected for both 2016 and 2024.

7.1.2 Spatial Boundary Data

Census tract shapefiles were obtained from the TIGER/Line Shapefiles provided by the U.S. Census Bureau. Census tracts were used as the primary spatial unit of analysis.

7.1.3 Crime and Environmental Data

Crime and park data were collected from Open Data Buffalo. Crime data included neighborhood crime counts, while environmental data included park locations and park area calculations.

7.1.4 Tax Assessment Data

Property assessment and tax data were obtained from the Buffalo Department of Assessment and Taxation.

7.2 Data Preprocessing

Several preprocessing steps were performed before analysis.

- All datasets were projected into a common coordinate system to ensure spatial consistency.
- Census tract shapefiles were clipped to the Buffalo city boundary.
- Demographic, tax, and environmental datasets were joined to census tract polygons using GEOID.
- Missing values and inconsistent records were checked and cleaned.

- Park areas were summarized within census tracts using ArcGIS to calculate the percentage of park coverage.
- Distance from each census tracts polygon center to downtown center of Buffalo was calculated using ArcGIS Euclidian distance tool.

7.3 Identifying Eligible Census Tracts:

The first methodological step was identifying census tracts vulnerable to gentrification. A binary eligibility filter was created using baseline 2016 socio-economic conditions.

A census tract was considered eligible for gentrification if:

- Median household income in 2016 was below the city-wide median income.
- Median home value in 2016 was below the city-wide median home value.

This was a necessary step because already wealthy neighborhoods experiencing increasing housing price do not necessarily represent gentrification. The eligibility filter isolated economically vulnerable neighborhoods where redevelopment pressures were more likely to produce displacement and demographic change.

7.4 Creating a Continuous Gentrification Score:

Instead of classifying neighborhoods as simply gentrified or not gentrified, this study developed a continuous gentrification intensity score.

Three economic indicators were standardized using Z-scores:

- Income change
- Home value change
- Rent change

$$z = \frac{x - \mu}{\sigma}$$

$\mu =$ Mean
$\sigma =$ Standard Deviation

The standardized values were combined into a composite gentrification score. The final score was then standardized again to produce a continuous gentrification intensity index.

Eligible census tracts were then divided into three categories:

- Low-intensity gentrification
- Medium-intensity gentrification
- High-intensity gentrification

This approach allowed the study to identify varying degrees of neighborhood transformation across Buffalo.

7.5 Statistical Analysis:

A multiple linear regression model was used to identify the socio-economic and spatial drivers of gentrification intensity.

The dependent variable is:

- Standardized gentrification intensity score

The independent variables included:

- Median income in 2016
- Population density in 2016
- Percent White population in 2016
- Crime count
- Housing age
- Average property tax in 2019
- Distance to downtown

- Percent Park coverage

The regression model evaluated which baseline neighborhood factor are significantly associated with later gentrification intensity.

7.6 Impact Analysis:

After classifying tracts into low-, medium-, and high-intensity categories, demographic and economic changes were summarized for each category.

The impact analysis examined:

- Home value increase
- Income increase
- Tax changes
- Changes in racial composition
- Changes in foreign-born population
- Changes in renter population
- Crime reduction

This analysis helped identify how neighborhood transformation differed across varying intensities of gentrification.

8. Result and Discussion:

8.1 Spatial Distribution of Gentrification:

The analysis identified 28 census tracts that were economically vulnerable and eligible for gentrification. These tracts were primarily concentrated in historically lower-income areas of Buffalo.

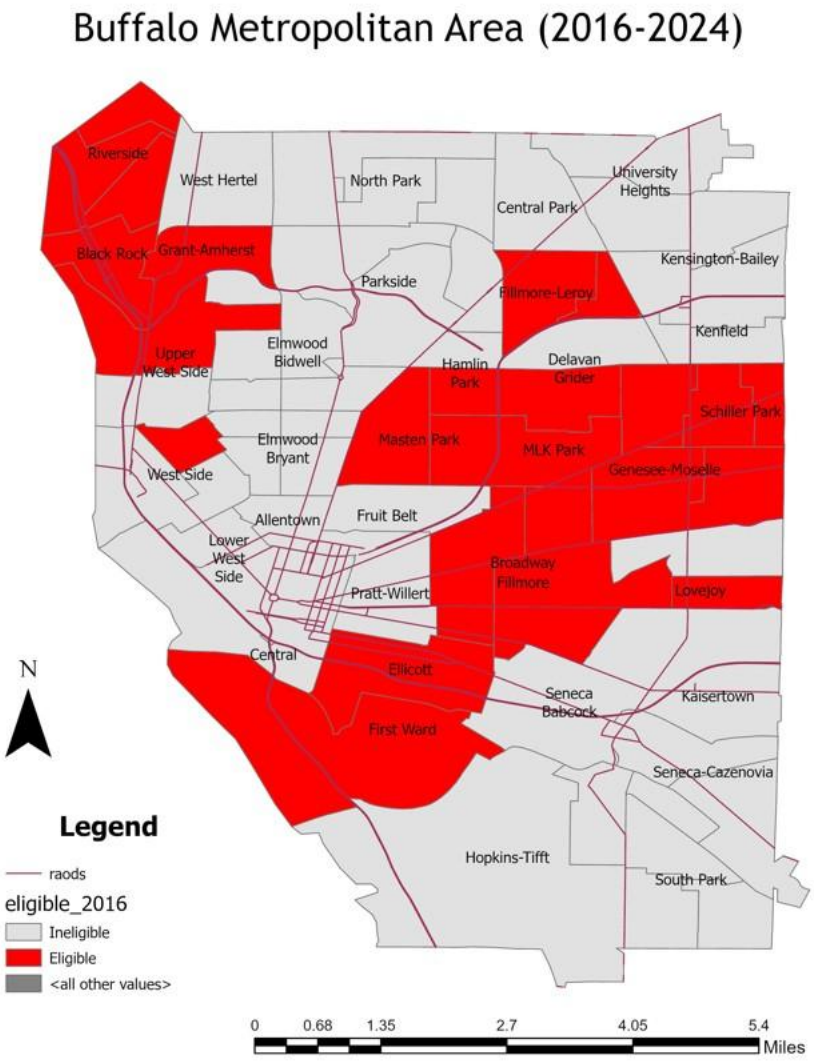


Fig: 1

The continuous gentrification score revealed substantial variation in neighborhood transformation intensity across the study area. Some neighborhoods experienced relatively limited economic change, while others underwent rapid increases in home values, rents, and household income.

High-intensity gentrification zones were generally concentrated in neighborhoods experiencing significant reinvestment and rapidly increasing property values. Medium-intensity areas appeared to function as transitional zones experiencing moderate redevelopment pressure, while low-intensity areas showed slower rates of neighborhood change.

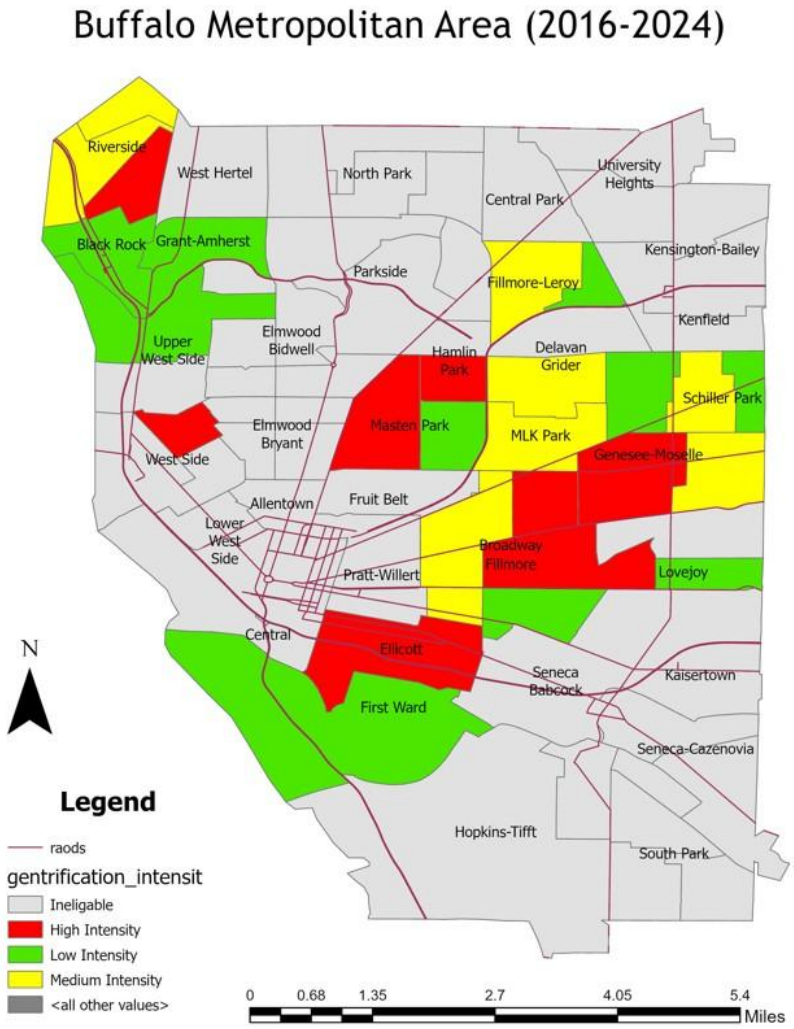


Fig: 2

8.2 Regression Results

The multiple linear regression model explained 56.4% of the variability in gentrification intensity.

Among all variables included in the model, only baseline median income was statistically significant at the 0.01 level. The coefficient for median income was negative, indicating that lower-income neighborhoods were more vulnerable to intense gentrification.

Population density, racial composition, crime count, housing age, property tax levels, park coverage, and distance to downtown were not statistically significant predictors.

These findings strongly support the Rent Gap Theory proposed by Smith (1979). The results suggest that gentrification in Buffalo is primarily driven by economic opportunity and undervalued land rather than by environmental amenities or neighborhood safety.

The regression findings also challenge the assumption that crime reduction or proximity to green space are major drivers of redevelopment in Buffalo. Investors and developers appear to prioritize low-cost neighborhoods with high future profit potential over other neighborhood characteristics.

```
Call:
lm(formula = gentrification_score_std ~ median_income_16 + pop_density_16 +
    pct_white_16 + crime_count + house_age_16 + avgtax2019 +
    dist_downtown + pct_park, data = data_eligible)

Residuals:
    Min      1Q  Median      3Q     Max
-2.0117 -0.3637  0.1220  0.4863  1.0109

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       2.030e+01  6.771e+01   0.300  0.76753
median_income_16 -2.094e-04  5.594e-05  -3.743  0.00138 **
pop_density_16    1.686e-04  1.571e-04   1.073  0.29665
pct_white_16     -1.277e+00  8.006e-01  -1.595  0.12717
crime_count       2.710e-03  2.843e-03   0.953  0.35248
house_age_16     -1.998e-01  8.806e-01  -0.227  0.82294
avgtax2019        1.287e-03  1.505e-03   0.855  0.40333
dist_downtown    -1.999e-01  1.289e-01  -1.551  0.13742
pct_park         -3.749e+00  4.262e+00  -0.880  0.38999
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.7872 on 19 degrees of freedom
Multiple R-squared:  0.564,     Adjusted R-squared:  0.3804
F-statistic: 3.072 on 8 and 19 DF,  p-value: 0.02127

>
```

8.3 Economic Impacts:

High-intensity tracts experienced the largest increases in housing values. Average home values increased by approximately 257.5% in these neighborhoods. Median household income also increased substantially, rising by approximately 133.1%(Table:1).

Property taxes also increased sharply. High-intensity neighborhoods experienced approximately 176.4% tax increases between 2024 and 2025(Table:1). These sudden tax increases may create additional financial pressure on long-term residents, including homeowners.

Gentrification Intensity	Home Value Change	Income Change	Tax Spike ('24–'25)
Low Intensity	+133.0%	+49.2%	+$28.61%
Medium Intensity	+161.6%	+79.3%	+$84.7%
High Intensity	+257.5%	+133.1%	+176.43%

Table:1

8.4 Demographic Change:

High-intensity gentrification areas experienced a decline in Black population shares alongside increases in White population shares. The White population increased by approximately 8.8%, while the Black population declined by approximately 16.5%(Table:2). At the same time, renter populations declined in medium- and high-intensity tracts.

Medium-intensity neighborhoods showed the strongest increases in Asian and foreign-born populations. The Asian population increased by approximately 10.9%, while the foreign-born population increased by approximately 7.5%(Table:2). These findings suggest that medium-intensity neighborhoods may function as transitional or buffer zones.

Gentrification Intensity	White Pop	Black Pop	Asian Pop	Foreign-Born	Renter Pop
Low Intensity	-3.0%	-4.2%	+5.2%	+3.8%	+0.0%
Medium Intensity	+0.9%	-15.6%	+10.9%	+7.5%	-4.0%
High Intensity	+8.8%	-16.5%	+2.8%	+1.5%	-4.6%

Table:2

8.5 Crime and Neighborhood Change:

Crime rates declined across all gentrification intensity categories. However, the largest crime reductions occurred in low-intensity neighborhoods rather than high-intensity areas. This finding further supports that Gentrification in Buffalo does not appear to be primarily driven by safer neighborhoods. Instead, redevelopment appears more strongly connected to economic opportunity and undervalued property markets, crime reduction is a consequence of neighborhood transformation and demographic change.

Gentrification Intensity	Crime Shift
Low Intensity	-22.1%
Medium Intensity	-10.5%

High Intensity	-20.2%

Table:3

The overall findings suggest that Buffalo's gentrification process closely follows patterns predicted by the Rent Gap Theory. The most economically vulnerable neighborhoods experienced the strongest redevelopment pressure and the greatest increases in housing values.

Although the result shows a demographic transition it cannot be concluded it's a displacement without further longitudinal study. Arrival of higher-income residents can improve neighborhood conditions, increase local business activity, and create job opportunities for long-term residents.

9. Conclusion

This project analyzed the spatial and socio-economic drivers of gentrification and its impacts in Buffalo between 2016 and 2024 using GIS and statistical analysis.

The study first identified economically vulnerable census tracts and then developed a continuous gentrification intensity score based on changes in income, housing values, and rent. Multiple regression analysis showed that baseline neighborhood poverty is the strongest predictor of later gentrification intensity.

The results strongly support the Rent Gap Theory by demonstrating that redevelopment in Buffalo is primarily associated with economic opportunity and undervalued land rather than crime reduction or environmental amenities.

10. Reference

1. Adelman, R. M., Balta Ozgen, A., & Rabii, W. (2019). Buffalo's West Side story: Migration, gentrification, and neighborhood change. City & Community, 18(3), 770–791. https://doi.org/10.1111/cico.12412
2. Çobangil, O., & Ateş, E. (2025). Airbnbfication and gentrification from the perspective of rent gap theory: The case of Beyoğlu, Istanbul. Urban Geography, 46(8), 1774–1800. https://doi.org/10.1080/02723638.2025.2481799
3. Coley, J., & Adelman, R. M. (2020). Gentrification in the "City of Good Neighbors": Race, class, and neighborhoods in Buffalo. Sociological Inquiry, 91(4), 824–848. https://doi.org/10.1111/soin.12387
4. Freeman, L. (2005). Displacement or succession? Residential mobility in gentrifying neighborhoods. Urban Affairs Review, 40(4), 463–491. https://doi.org/10.1177/1078087404273341
5. Hohle, R. (2024). A grassroots alternative to urban shrinkage? A comparative analysis of place reputational remaking in Buffalo and Cleveland. Urban Affairs Review, 60(6), 1734–1771. https://doi.org/10.1177/10780874241237435
6. Lees, L., Slater, T., & Wyly, E. (2008). Gentrification. Routledge.
7. Pichardo, C. M., Chambers, E. C., Sanchez-Johnsen, L. A. P., Pichardo, M. S., Gallo, L., Talavera, G. A., Pirzada, A., Roy, A., Castañeda, S. F., Durazo-Arvizu, R. A., Perreira, K. M., Teng, Y., Rodriguez, C. B., Allison, M., Carlson, J. A., Daviglus, M. L., & Plascak, J. J. (2023). Association of census-tract level gentrification and income inequality with 6-year incidence of metabolic syndrome in the Hispanic Community Health Study/Study of

Latinos, an epidemiologic cohort study. Social Science & Medicine, 336, Article 116222. https://doi.org/10.1016/j.socscimed.2023.116222

8. Silverman, R. M., Taylor, H. L., Yin, L., Miller, C., & Buggs, P. (2018). Perceptions of residential displacement and grassroots resistance to anchor driven encroachment in Buffalo, NY. Urbanities, 8(2), 79–86.
9. Silverman, R. M., Taylor, H. L., Jr., Yin, L., Miller, C., & Buggs, P. (2019). There goes our family friendly neighborhood: Residents' perceptions of institutionally driven inner-city revitalization in Buffalo, NY. Journal of Community Practice, 27(2), 168–187. https://doi.org/10.1080/10705422.2019.1616642
10. Smith, N. (1979). Toward a theory of gentrification: A back to the city movement by capital, not people. Journal of the American Planning Association, 45(4), 538–548. https://doi.org/10.1080/01944367908977002
11. Swanstrom, T., Webber, H., & Metzger, M. (2015, April 2–3). Rebound neighborhoods in older industrial cities: The case of St. Louis [Paper presentation]. Federal Reserve System Community Development Conference, Washington, DC, United States.
12. Vergara, J. V., Rodriguez, M. Y., Dohler, E., Phillips, J., Villodas, M., Wilson, A. B., & Joseph, K. (n.d.). Promises and pitfalls of a new early warning system for gentrification in Buffalo, NY [Unpublished manuscript]. Department of Computer Science and Engineering, University at Buffalo.
13. Zhang, Z., & Barr, A. (2024). Gentrification and crime in Buffalo, New York. PLOS ONE, 19(6), Article e0302832. https://doi.org/10.1371/journal.pone.0302832